24 Monsanto Plant Disaster. Content warning: #explosion

Why Are We Learning This?

Chemical reactor design is not only about maximizing conversion and production.

It is also about:

- Ensuring reactor safety
- Managing risk
- Preventing catastrophic failure

Reactor accidents provide opportunities to understand how:

- Reactor design
- Heat transfer
- Reaction kinetics
- Safety systems

work together to prevent catastrophic failure.

Learning Objectives

By the end of this lesson, you should be able to:

✅ Explain the causes of thermal runaway

✅ Describe the role of a heat exchanger in reactor safety

✅ Explain how rupture disks provide emergency protection

✅ Compare heat generation and heat removal

✅ Develop batch reactor mole and energy balances

✅ Evaluate the impact of operating decisions on reactor safety

✅ Propose safer reactor operating procedures

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions

↓
►Reactor Safety◄
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

This problem is taken from Elements of Chemical Reaction Engineering, 5th Edition, by H. Scott Fogler:

A serious accident occurred at the Monsanto plant in Sauget, IL on August 8, 1969 at 12:18am. The blast was hear as far as 10 miles away in Belleville, IL. Th explosion occurred in a batch reactor that was used to produce nitroaniline (A) from ammonia (B) and o-nitrochlorobenzene (C):

A + 2B --> C + D

Where D = NH4Cl. The reaction was carried out in water solvent (W).

This reaction was normally carried out isothermally at 175C and about 500 psi. A heat exchanger carrying water at 25C was used to maintain isothermal operation.

The temperature was controlled by varying the flowrate of water through the heat exchanger. At slower flowrates, the water would get warmer as it flowed through the heat exchanger. At higher flowrates, the water temperature remained constant at 25C. Maximum cooling hence occurred at higher flowrates.

Isothermal operation was in part a safety feature of this reactor. The reaction is pretty exothermic (heat of reaction = -590 kcal/mol); further it has a pretty low activation energy (11.2 kcal/mol) meaning that the rate increases rapidly with temperature.

Further, at 300C, nitroaniline decomposes into noncondensable gases such as CO, N2, and NO2. Production of noncondensable gases leads to high pressure buildup and hence must be avoided.

The main safety feature on the Monsanto reactor was a rupture disk, which was designed to burst when the pressure in the reactor exceeded 700 psi. Rupture of the disk would decrease the pressure, which would cause the water in the reactor to vaporize, which would remove some of the heat generated by the reaction and hence cool the reactor. Further, removal of steam through the relief valve would remove significant heat from the reactor (Fogler estimates 45,000 kcal/min) and prevent explosion.

Maybe because of this, or maybe for another reason, the reactor was not equipped with a backup cooling system. Instead, when the heat exchanger would fail, an operator would run out and get it running again. As long as they did this within 10 minutes of heat exchanger failure, there hadn't been any problems.

The day the reactor exploded, several things had gone wrong.

1. The reactor was designed for a charge of 3.17 kmol A and 43 kmol B along with 103.7 kmol H_2O. However, the day the reactor exploded, it had been charged with 9.044 kmol A, 33 kmol B, and 103.7 kmol H_2O. It is speculated that Monsanto was interested in increasing production so they increased the charge to the reactor.
2. The day the reactor exploded, the heat exchanger failed, and the technician wasn't able to get it working until 10 minutes later.
3. The rupture disk also failed.

We are going to model this reaction to show that had any *one* of those things not gone wrong, the reactor wouldn't have exploded.

First, let's model normal reactor operation. The following data and information was taken from Fogler:

- Assume the reaction is elementary
- k = 0.00017 m^3/kmol*min at 188C
- E = 11.3 kcal/mol
- System volume: 3.26 m^3 for "designed" charge, 5.12 m^3 for overcharge
- Heat of reaction = -590 kcal/mol

- C_{pA} = 0.040 kcal/mol*K
- C_{pW} = 0.018 kcal/mol*K
- C_{pB} = 0.00838 kcal/mol*K
- UA = 35.85 kcal/min*K
- The maximum cooling rate is achieved when T_a = 298 K

1. Determine an expression for the maximum cooling rate, $Q_{r,max}$ for this reactor.
2. Develop the mole balance for isothermal operation. Use it to determine X and Q_g after 45 minutes of operation, which is when the heat exchanger failed. Do this for the intended charge of A as well as the overcharge of A. Compare with the maximum value of Q_r.
3. Now, model adiabatic operation for 10 minutes. What are X, T, and Q_g for the original and new charge now? Compare again with the maximum value of Q_r.
4. Vary the following parameters in order to develop a set of safety criteria that would ensure the reactor would never explode. (We will assume that at least one backup heat exchanger was added to this reactor, that the rupture disk will be tested regularly, and that multiple types of rupture disks have been added. Let's still design for the possibility that everything fails at once.)
   a. Vary the initial charges of A, B, W assuming the heat exchanger fails for 10 minutes after 45 minutes of operation
   b. Vary the time at which the heat exchanger fails, assuming it fails for 10 minutes and that the reactor is charged as designed
   c. Vary the amount of time that the heat exchanger is down for, assuming it fails after 45 minutes and that the reactor is charged as designed
   d. Vary the time at which the heat exchanger fails, assuming it fails for 10 minutes and that the reactor is overcharged
   e. Vary the amount of time that the heat exchanger is down for, assuming it fails after 45 minutes and that the reactor is overcharged

What is the maximum charge to the reactor that ensures an explosion will *never* happen?

Normal Reactor Operation

The reactor normally operated Isothermally at approximately:

Temperature = ______________

Pressure = _________________

A heat exchanger carrying water at: ______________________________ was used to maintain reactor temperature.

Why Was Isothermal Operation Important?

The reaction was (circle one):

Exothermic or Endothermic

Heat of reaction:

$\Delta H_{rxn} =$ __________________________

The activation energy was relatively (circle one): High or Low

Therefore:

Temperature Increase

↓

____

↓

____

Additional Hazard

At approximately 300°C the product decomposes to form:

- CO

- N_2
- NO_2

These are noncondensable gases.

Why is that dangerous?

Noncondensable gases accumulate in the reactor. Pressure increases rapidly. High pressure can lead to equipment failure or reactor explosion if not relieved.

Reactor Safety System

The primary safety device on the reactor was a Rupture Disk designed to burst when pressure exceeded 700 psi.

Why Was the Rupture Disk Important?

Complete the sequence:

Pressure Increases
↓

____
↓
Pressure Drops
↓

____
↓
Energy Removed
↓
Temperature Drops

Fogler estimates that steam removal could remove approximately:

______________________________ kcal/min.

What Actually Went Wrong?

Several things failed simultaneously.

Problem #1

____

Problem #2

____

Problem #3

____

Intended versus Actual Charge

Intended Reactor Charge

A: 3.17 kmol

B: 43 kmol

Water: 103.7 kmol

Actual Reactor Charge

A: 9.044 kmol

B: 33 kmol

Water: 103.7 kmol

Engineering Question

Why might increasing the initial amount of A increase the likelihood of runaway?

More Reactant
↓
More Heat Generated
↓
Higher Reactor Temperature
↓
Greater Risk

Heat Generation versus Heat Removal

The reactor is safe when: Heat Removal > Heat Generation

When the opposite occurs: Heat Generation > Heat Removal

then:

Temperature Increases --> Reaction Rate Increases --> Thermal Runaway May Occur

Modeling Approach

We will investigate:

Part 1: Normal operation for: 45 minutes

Questions:

- What is X?
- What is G(T)?

Part 2

Adiabatic operation for: 10 minutes

after heat-exchanger failure.

Questions:

- What is X?

- What is T?
- What is $\dot{Q}_g$?

Reactor Volume

Designed Charge: $V =$ **____________________________** m³

Overcharged Reactor: $V =$ **________________________** m³

Important Physical Properties

Rate constant: $k =$ ____________________ at $T =$ ___________________________

Activation energy: $E =$

Heat of reaction: $\Delta H_{rxn} =$

Safety Criteria Investigation

We will investigate how reactor safety changes when we vary: **Charge Amount, Failure Time**

Cause-and-Effect Diagram

Fill in the missing boxes:

Heat Exchanger Failure
↓

____
↓
Temperature Rises
↓

____

↓

Reaction Rate Increases

↓

____

↓

Thermal Runaway

Reflection Questions

Question 1: What aspect of the reactor design appears most important for preventing thermal runaway?

Question 2: Why was the rupture disk more than just a pressure-relief device?

Question 3: Why was the increased charge potentially dangerous?

Question 4: Why might multiple independent safety systems be preferable to a single safety system?

Question 5: If you were redesigning this reactor today, what additional safety feature would you add?

Most Important Idea

Heat Generation vs. Heat Removal

Safe operation requires: Heat Removal ≥ Heat Generation

When multiple safety systems fail simultaneously: Heat Generation > Heat Removal

which can lead to Thermal Runaway and ultimately Explosion.